Manipulation of Voters in Competitive Election Campaigns

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Abstract

We provide a game-theoretical model of manipulative election campaigns with two political candidates and a continuum of Bayesian voters. Voters are uncertain about candidates' policy platforms. Candidates take unobservable, costly actions to manipulate public opinion about their positions. We show that if the candidates differ in campaigning efficiency, and voters receive the biased campaign messages with some noise, then the cost-efficient candidate wins the election even if she is more distant from the electorate than her opponent is. Our paper offers a novel informational justification for imposing limits on campaign spending and encouraging diversity in the supply of political information.

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1 Introduction

Politicians often face a trade-off between gaining popular support and pursuing their own agenda. The less aligned their policy preferences are with that of the majority, the more incentive they have to manipulate public opinion. Especially during election campaigns, candidates may choose to misrepresent their preferences in order to appeal to a large audience. Aware of politicians' incentives, rational voters face an inference problem when evaluating political messages. Their ability to overcome this problem depends, among other things, on the characteristics of the communication environment mediating these messages. In this paper, we offer a model of manipulative election campaigns where the structure of communication allows candidates to undermine voters' ability to infer true candidate positions.

Despite the common wisdom and the evidence that election campaigns can mislead voters (Ramsay et al. 2010, Cassino and Wooley 2011, DellaVigna and Kaplan 2007, Durante and Knight 2012, Puglisi and Snyder 2011), political campaigns are often modeled as entirely informative activities in the literature on electoral competition (Coate 2004a, Gul and Pesendorfer 2012). Although there is a wide variety of models studying the limits with respect to the quality of information that voters receive, the equilibrium beliefs that voters form about the candidates usually remain unbiased. This is because the information structure adopted in these models allows rational voters to discount biased messages to the full extent (e.g., Wittman 1989), leaving no incentive for politicians to provide false information in equilibrium (Banks 1990, Aragones and Neeman 2000, Martinelli 2001). In existing models, even when voters end up holding biased beliefs about the true state of the world, these beliefs are supported in equilibrium only when voters have bounded rationality or when they already had biased beliefs (Heidhues and Lagerlof 2003) prior to the election campaigns.[1] In contrast to existing theoretical approaches to political campaigns, in this paper, we argue that even when voters are rational and do not hold biased prior beliefs, under a non-transparent social communication environment, deceptive political propaganda can become an equilibrium outcome.

We propose a two-candidate electoral competition model in which the candidates know voters' most-preferred policy, but voters are uncertain about the candidates' positions. Under this informational asymmetry, we provide sufficient conditions for political candidates to successfully bias the beliefs of Bayesian voters about their policy

[1] For a comprehensive discussion of models with bounded rationality, see Caplan (2006).

positions and to influence election outcomes. We model information manipulation as directed disinformation on a unidimensional policy space: Candidates try to manipulate voters' beliefs about their relative proximity to voters through manipulative and costly actions that shift an otherwise unbiased signal. Candidates' objective is to maximize the votes they get, and the outcome of the competition is determined by majority voting among a large set of voters. The campaigning efficiencies of the politicians are allowed to differ, and this is captured by asymmetric variable costs of manipulative actions.[2]

Using this general framework, we analyze when and how information manipulation in election campaigns leads to a misinformed electorate. In addition to the standard assumption of information asymmetry between politicians and the electorate, we argue that, within the framework of our paper, there are three key assumptions without which manipulation cannot succeed: (i) asymmetry between the candidates in terms of campaigning efficiency (costs); (ii) an idiosyncratic noise –or bias– in campaign signals received by the electorate; and (iii) the lack of perfect communication among voters. We show that when these conditions hold, the candidates tailor their manipulative actions in such a way that voters, who cannot perfectly distinguish idiosyncratic noise from the net manipulation in campaign messages, encounter an inference problem.

The idiosyncratic noise in campaign signals can be broadly interpreted as a byproduct of a social communication environment in which information received by voters exhibits diversity due to different social habits of information collection and processing, or due to differential access to media channels and social communication networks. Studying such a social structure in detail is beyond the scope of this paper. Hence, we simply represent this structure as an exogenous random process that converts social actions by influential players into noisy signals to be received by the electorate.[3]

Our analysis suggests that unregulated electoral competition in a representative democracy may lead to socially inefficient election outcomes by biasing the electorate's beliefs in favor of the candidate who is more efficient in manipulation, but who does not necessarily stand closer to voters. We prove that under manipulative election cam-

[2]Asymmetric costs may stem from unequal access to campaigning resources and technology, which, among other things, may depend on how their policy preferences and personal characteristics appeal to special interest groups or whether they come from an incumbent political party with superior tools of information manipulation at their disposal.

[3]Alternatively, this heterogeneity among voters can also be interpreted as the byproduct of ideological biases in voters' beliefs about the candidates, relative to that of the median voter. This can be a reasonable interpretation as long as the degree of relative biases in beliefs is unknown to each voter, while its distribution is common knowledge.

paigns, even when the cost-efficient candidate is ex-post an inferior choice for voters, this candidate can win the election as long as she does not have too great a positional disadvantage on the policy space vis-à-vis her opponent. Although this adverse effect on election outcomes vanishes when campaigning costs become symmetric, there is a second source of inefficiency in manipulative election campaigns that nonetheless remains. Even when the campaigning costs of the candidates are identical, the war of manipulation between the candidates continues, resulting in wasteful campaign expenditures.

We then present several policy-relevant results and extensions to our benchmark model. First, we analyze how a change in the gap between the candidates' cost efficiency influences the extent of net bias in electoral messages. Confirming what Gul and Pesendorfer (2012) have found, we demonstrate that as the difference in manipulative effectiveness decreases, so does the net bias in campaign messages.

Next, we present a slight modification to our benchmark model to illustrate the beneficial effect of limiting campaign spending in the presence of special-interest groups. We prove that a planner that tries to minimize the systematic bias in voters' beliefs would limit the amount of allowable contributions such that candidates will no longer have an incentive to engage in manipulation.

Third, we analyze how a change in the precision of campaign messages affects the candidate strategies and vote shares. Numerical simulations demonstrate how higher signal precision renders manipulation more effective and increases the vote share of the cost-efficient candidate over a certain range of relative candidate positions.

Finally, we discuss two extensions to our benchmark model. First we explore whether our main conclusions about the effectiveness of manipulative campaigns can survive when there is a finite number of voters. In this setting, being pivotal is a probable event. Therefore each individual votes strategically and forms her beliefs as if she is the pivotal voter. We argue that there exists an equilibrium in monotonic strategies. In this equilibrium, we numerically show that the net manipulation is positive, and strategic voting does not eliminate manipulation. Second, we demonstrate that manipulation remains effective in equilibrium when voters are heterogenous with respect to their political preferences. Allowing for heterogeneity in voters' bliss points – in addition to the heterogeneity in private signals – does not eliminate the systematic bias in voter beliefs.

The paper is organized as follows: Section 2 briefly reviews the related literature.

Section 3 describes the model while Section 4 studies the electoral equilibria when manipulation is feasible and when it is not. Section 5 is devoted to a discussion of the sufficient conditions for manipulative campaigns to succeed in equilibrium. We end this section with an illustrative example demonstrating the nature of voters' inference problem. Section 6 presents numerical simulations of the model with regard to the extent of cost asymmetry between the candidates and the degree of informational heterogeneity among voters. Potential policy implications of our model are also discussed in this section. Section 7 provides some extensions to the benchmark model and demonstrates robustness with respect to strategic voting and heterogeneity among voter preferences. Finally, Section 8 concludes. Omitted proofs are collected in the Appendix.

2 Related Literature

Information manipulation is an outcome of signaling games, where signal receivers' equilibrium beliefs mislead them into strategically inferior actions. This can happen when signal receivers hold biased prior beliefs, and the signals they receive do not correct this bias. Cummins and Nyman (2005) study such a case in which better-informed investing firms choose to send signals that confirm the biased priors of less-informed investors. Gentzkow and Shapiro (2006) prove a similar result for media firms. Similarly, Heidhues and Lagerlöf (2003) show that if voters hold biased prior beliefs, political candidates may have an incentive to send biased signals to voters that confirm these priors. None of these papers considers the heterogeneity among signal receivers, and all of them assume that the average prior belief of receivers is biased.

In the context of politics, there are a few studies featuring information manipulation by a signal sender and rational signal receivers (e.g. Ekmekci 2009 and Edmond 2013). Edmond (2013) studies the interaction between a manipulative dictator and a continuum of heterogeneous citizens within the framework of global games. Ekmekci (2009) studies the manipulative incentives of an expert to mobilize voters to coordinate around a candidate she favors. They both prove that information manipulation is possible in equilibrium and has a real effect on signal receivers' actions. In both papers, there is one manipulator; signal receivers' incentives are coordinated, and the source of the heterogeneity among agents is an idiosyncratic noise added to the manipulator's messages.

Our theoretical framework differs from these two studies in various aspects. First, our model features two competing manipulators, instead of a single manipulator. Second, we do not assume any strategic complementarity on the side of the voters. Voters care only about the policy outcome and not about the size of the support that the winning candidate gets. Therefore, a candidate does not become more attractive for a voter just because more of the remaining voters choose to support her. Hence, in our benchmark model, there is no scope for *strategic coordination* among signal receivers. This is different from the strategic environment in Edmond (2013), in which a citizen's relative payoff from supporting a revolution against a dictator (the manipulator) – instead of remaining passive – depends on how other citizens behave. Consequently, the models in Ekmekci (2009) and Edmond (2013) do not differentiate between *strategic coordination failure* and *informational coordination failure*. In contrast, we analyze a political setting in which incentives for strategic coordination are not present, and show that the success of political manipulation solely stems from an informational coordination failure.

Our results have different policy implications regarding the regulation of election campaigns than most other studies in the literature. Theoretical explanations for why campaign spending should not be limited commonly assume that campaign messages do not contain misleading information. Coate (2004a) studies such informative campaigns and addresses the question of how campaign contributions of interest groups are related to the election strategies of political parties. He proves that parties randomize between extremist and moderate candidates to cope with the trade-off between attracting the interest groups by supporting an extremist candidate and winning the election by supporting a moderate candidate. Gul and Pesendorfer (2012) propose another approach for informative election campaigns. In their continuous-time model, candidates prefer to provide more information if the information will favor them. They show that when spending is limited, the amount of revealed information is reduced, and, thus, voters become worse-off. Both of these papers suggest that lowering limits on campaign spending is welfare-improving. Our model implies the opposite, as we assume that campaigns are manipulative.

There are a few studies which, like our paper, conclude that limiting campaign contributions can improve welfare (Grossman and Helpman, 2001; Prat, 2002a, 2002b; and Coate, 2004b). However, in existing studies, campaigns do not bias voter beliefs. In contrast, we model political advertising as costly signal jamming, which, in equilibrium,

leads to posterior voter beliefs that are biased towards the cost-efficient candidate. As a result, limiting campaign contributions can improve welfare by reducing the ex-ante systematic bias in voters' beliefs. To the best of our knowledge, this type of a justification for campaign spending limits, which relies on the systematic bias in voters' beliefs, has not been formally analyzed before.

Finally, the manipulation-based argument for spending limits, that we develop here, has policy implications for campaign regulation which potentially differ from those implied by the existing models. For instance, in Coate (2004b), replacing "favor financed" campaign spending with a matching tax-financed public subsidy can offset the adverse informational effects of a limit on campaign contributions, further increasing the likelihood that a qualified candidate will come to power. In our model, however, even if we had allowed for informative advertising and campaign contributions from interest groups, changing the source of funding from private to public contributions by itself would not lower the net manipulation in campaign signals. This is because public financing cannot be conditioned on truthful advertising unless regulatory agencies can distinguish manipulative advertising from non-manipulative advertising.

3 The Model

In this section, we lay out a model of election campaigns with two political candidates and a large mass of voters. The model features an asymmetric information structure. The candidates are fully informed about preferences of voters, while voters are uncertain about candidates' political positions. Voters observe only a noisy signal about candidates' relative proximity to their common bliss point and vote for the candidate that they believe stands closer to them. The key aspect of the model is a "war of disinformation" between the candidates whereby they can distort the signals received by voters via costly campaign messages. One of the candidates is more cost-efficient in manipulation than the other candidate, so that the net distortion in signals favors her whenever there is an incentive for manipulation. Since voters understand the manipulative nature of the campaigns and are fully rational, they correctly anticipate the net direction of the distortion and act accordingly. However, voters, on average, underestimate the magnitude of this distortion as long as an idiosyncratic noise blurs the communication between voters and the candidates.

Consider an electoral competition between two candidates indexed by $j \in P =$

$\{I, C\}$ and a continuum of voters with a unit mass indexed by $i \in [0, 1]$.[4] Each candidate holds an exogenously given political position $\omega_j \in \mathbb{R}$.[5] Voters hold a common political position given by $\bar{\omega} \in \mathbb{R}$, which we call voters' bliss point.[6] We define $\theta \in \mathbb{R}$ as an index comparing the political distances of candidates to voters' common bliss point $\bar{\omega}$. That is,

$$\theta = |\omega_C - \bar{\omega}| - |\omega_I - \bar{\omega}| + \frac{1}{2},$$

where 1/2 is an arbitrary rescaling parameter. Since $\theta > 1/2$ if and only if I stands closer to voters than C, we will refer to θ as the relative proximity of candidate I.

Information structure and manipulative actions

Voters' bliss point $\bar{\omega}$ is common knowledge among all players. While the candidate positions $\omega_I, \omega_C \in \mathbb{R}$ are common knowledge among the candidates, voters do not observe these positions.[7] Therefore, θ is a hidden state variable for voters, whereas candidates I and C are able to condition their actions on the realization of this state variable. In particular, given θ, candidates simultaneously choose manipulative actions a_I,$a_C \geq 0$ to signal that they are closer to voters than what is suggested by θ. These manipulative actions can take the form of political statements, advertisements or other political actions with an intention to mislead voters.[8]

Manipulative actions shift the state variable θ in opposite directions, resulting in an intermediate state variable $\theta + a_I - a_C$. However, the electorate does not observe this variable directly due to its heterogeneous relationship with the communication environment that interferes with the campaign signals. Formally, they receive a private

[4]Given the setup of the strategic interactions that we consider in the model, the assumption of a continuum of voters is not critical. Even if we assumed that cardinality of voters is finite, without changing the other assumptions, our arguments about the effectiveness of manipulation would continue to hold. To keep our model tractable, we abstract from such considerations.

[5]If we allowed candidates to choose their policy platforms prior to campaigning, both candidates would choose $\bar{\omega}$ to minimize the costs of manipulation, unless there are other reasons – not modeled here – pulling them toward extreme policy positions. If voters continue to believe that policy platforms are exogenous, then there would still be manipulation in equilibrium.

[6]Informational heterogeneity among voters will be introduced when we discuss the information structure characterizing the election campaign. We analyze the impact of heterogeneity in preferences in Section 7.2

[7]This assumption is essential to capture the idea that voters and candidates are asymmetrically informed. Without such asymmetry, there would not be any incentive to manipulate voters' beliefs.

[8]Note that this broad conceptualization of information manipulation includes actions that disparage the opponent, such as attack ads, as well as any statement that paints an overly optimistic image of the manipulator. However, we do not make any distinction between these two forms of manipulation.

signal $x_i := \theta + a_I - a_C + \varepsilon_i$, where $\varepsilon_i \in \mathbb{R}$ is the random noise. $\{\varepsilon_i\}_{i\in[0,1]}$ are assumed to be i.i.d across the voters and have a normal distribution with zero mean and precision $\alpha > 0$ (i.e., variance α^{-1}).

Voters have uninformative priors about θ.[9] Therefore x_i is the only information that voter i can use to make a decision. This information may be biased due to both the net manipulation $\bar{a} := a_I - a_C$ and the idiosyncratic noise ε_i, which can also be interpreted as a individual bias. In the presence of idiosyncratic noise, voters cannot identify the location of their private signals in the population distribution unless there is perfect communication among them. Hence, we assume that voters do not communicate with each other instead of modeling an imperfect communication structure.[10]

Preferences and campaign costs

Each voter prefers the candidate, whose policy position is closer to $\bar{\omega}$. More formally, voter $i \in [0,1]$ supports C if and only if her posterior belief conditional on her private signal x_i satisfies $P(\theta \geq 1/2 \mid x_i) \leq 1/2$. This voting decision is described by a binary voting function $s_i(\cdot) : \mathbb{R} \to \{0,1\}$ such that $s_i(x_i) = 1$ if and only if i supports candidate C. After all voters simultaneously vote, the masses of supporters for both parties are determined. We denote the mass of supporters of candidate C by $S(a_I,\, a_C;\, \theta)$, which is equal to C's vote share and given by

$$S(a_I, a_C; \theta) \equiv \int_0^1 s_i(x_i) di. \tag{1}$$

For a candidate or political party that expects to run in an election only once, the most natural goal would be to win the election. However, many times, maximizing vote share would be a more realistic goal, especially when a larger vote share leads to higher legislative power in the parliament or when it helps the party raise more resources from interest groups. Candidates try to maximize their vote share subject to

[9]Technically, voters' priors follow an improper uniform distribution, which is the limiting distribution of a sequence of symmetric uniform distributions, whose bounds diverge to infinity. We abstain from imposing specific priors that are biased towards one or the other candidate to focus on how information manipulation affects voting behavior of voters who were neutral towards both candidates prior to the election campaign.

[10]Our analysis would not be qualitatively affected if we instead assumed imperfect communication. To locate their private signals precisely, voters would need to know the mean of $\{x_i\}_{i\in[0,1]}$. Even if each voter asks around to learn about the signals others have received, she cannot estimate the population mean without noise unless each voter can perfectly communicate with all the rest of the electorate.

the costs of manipulation.[11] Candidates' payoffs are given by

$$u_C(a_C, a_I; \theta) = 2S(a_I, a_C; \theta) - 1 - H_C(a_C)$$
$$u_I(a_I, a_C; \theta) = 1 - 2S(a_I, a_C; \theta) - H_I(a_I),$$

where $H_j : \mathbb{R}_+ \to \mathbb{R}_+$, for $j \in P$ govern the costs associated with manipulative actions.[12] For clarity of exposition, we adopt a specific form for the cost functions and impose the following assumption:

Assumption 1 *For each $j \in P$, $H_j(x) = \frac{1}{2}k_j x^2$ and $k_C > k_I \geq \left(\sqrt{\frac{2}{e\pi}}\right)\alpha \approx 0.484\alpha$.*

Quadratic cost functions imply that, as the level of campaigning increases, it becomes harder for candidates to send additional messages to the public. The assumption that $k_I < k_C$ formalizes the cost-advantage of candidate I. At any given level of manipulation, I pays a lower cost than C for each additional message. Due to her cost advantage, we will refer to candidate I also as the incumbent and to candidate C as the challenger, although this labeling does not imply any other difference between them. The second inequality, on the other hand, imposes a lower bound for the cost parameters to ensure the uniqueness of the equilibrium that will be defined below.[13] This property implies that the marginal costs of manipulation rise sufficiently fast. Cost functions are also common knowledge, but we assume that the cost levels associated with manipulative actions of political candidates are not observable by voters.[14]

[11] If, instead, we assumed that candidates care only about winning the election, our theoretical results would not be qualitatively affected. In that case, the manipulation game between the candidates would lead to corner outcomes where a candidate would choose zero manipulation if she is bound to lose the election, while the other candidate would engage in manipulation but still would not receive all the votes.

[12] Since the paper's focus is on the role of information manipulation, we assume that the level of campaigning effort by candidates is completely captured by manipulative actions a_j.

[13] As will be seen in the proof of Theorem 1, the equilibrium is characterized by an equation that is a linear combination of first derivatives of cost functions and the standard normal probability density function $\phi(\cdot)$. Non-invertibility of $\phi(\cdot)$ on its full support allows for multiple solutions to the equation, which implies multiplicity of equilibria. The condition that we impose on the second derivatives of the cost functions (k_I and k_C) ensures uniqueness of the equilibrium.

[14] This is a simplifying assumption that is consistent with the idea that voters may not know how campaign budgets are allocated between activities aiming to deceive voters about candidate qualities, on the one hand, and truly informative campaign activities, from which our model abstracts, on the other. Qualitatively, our results would go through as long as we assume that manipulative campaign spending by candidates is only imperfectly observed.

The sequence of events

Figure 1 describes the sequence of events in the election game. In the first stage, nature determines the relative proximity θ of candidate I to voters vis-à-vis candidate C, and each candidate strategically chooses her manipulative action. In its isolated form, the first stage of the election game is a simultaneous-move, two-person, normal-form game with strategic substitutes.

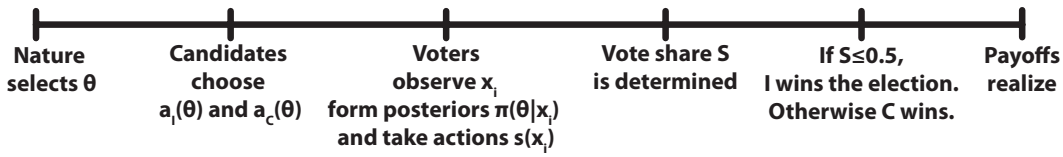

Figure 1: Timing of events in the election game

In the second stage, voters form their beliefs about θ conditional on their private signal x_i, which is jointly determined by the hidden state variable θ, manipulation actions of the first stage game and the random draws of individual bias ε_i. Given their beliefs, they vote for the candidate who they think is more likely to stand closer to their bliss point $\bar{\omega}$. Resulting vote shares determine the winner of the election and the associated payoffs for all players. Candidate I wins the election if and only if $S(a_I, a_C; \theta) \leq 1/2$, and C wins otherwise. In the next section, we define the equilibrium of the electoral game and present the main theoretical results of the paper.

4 Electoral Equilibrium

We employ the perfect Bayesian equilibrium (PBE) concept, which requires that each candidate develops a separate strategy in response to each possible voting strategy by the electorate, which is consistent with her expectations, and chooses the best action given her opponent's action. On the voters' side, the PBE rules out any belief about candidates' actions that contradict the information that voters have. Therefore, in equilibrium, voter beliefs and actions should be consistent with the fact that the incumbent has access to a superior manipulation technology and will choose a higher manipulation level than the challenger as long as candidates have an incentive to manipulate. The following definition lays out the conditions characterizing the electoral equilibrium:

Definition 1 (Electoral Equilibrium) *An electoral equilibrium consists of voter posterior beliefs $\pi(\cdot)$, voting behavior $s(\cdot)$, mass of supporters of the challenger $S(\cdot,\cdot;\cdot)$ and hidden actions $a_I(\cdot)$, $a_C(\cdot)$ such that $\forall i \in [0,1]$, $\theta, x_i \in \mathbb{R}$,*

$$\pi(\theta|x_i) = \frac{f(x_i|\theta, a_I(\theta), a_C(\theta))}{\int_{-\infty}^{\infty} f(x_i|\theta, a_I(\theta), a_C(\theta))d\theta},$$

$$s(x_i) = \begin{cases} 1, & \text{if } \int_{-\infty}^{1/2} \pi(\theta|x_i)d\theta \geq \frac{1}{2} \\ 0, & \text{if otherwise,} \end{cases}$$

$$S(a_I(\theta), a_C(\theta); \theta) = \int_{-\infty}^{\infty} s(x_i) f(x_i|\theta, a_I(\theta), a_C(\theta))dx_i,$$

$$a_I(\theta) \in arg \max_{a \geq 0} \{1 - 2S(a, a_C(\theta); \theta) - H_I(a)\}, \quad \text{and}$$

$$a_C(\theta) \in arg \max_{a \geq 0} \{2S(a_I(\theta), a; \theta) - 1 - H_C(a))\}$$

In the first expression above, $f(x_i|\theta, a_I(\theta), a_C(\theta))$ is the probability distribution function of private signals x_i conditional on θ and the manipulative actions of the candidates in response to θ. Note that the domain of the p.d.f. $f(.|.,.,.)$ is the set of private signals and not the set of relative proximity parameters, θ. Thus, the denominator in the first expression is not necessarily equal to unity.

Because of their simple structure and intuitive appeal, we will concentrate on equilibria in pure strategies where voters use monotonic voting rules. Under a monotonic strategy, a voter will support the incumbent candidate if and only if her private signal is above a certain threshold.[15] Formally, a monotonic strategy entails a threshold $\hat{x} \in \mathbb{R}$ such that $s(x_i) = 1 \Leftrightarrow x_i > \hat{x}$.

To highlight the impact of manipulative campaigns on aggregate voting behavior, we first analyze the electoral equilibrium without manipulation as a benchmark case. The following proposition shows that when the candidates cannot engage in manipulation —i.e., $\forall i \in [0,1]$, $x_i = \theta + \varepsilon_i$— the candidate who stands closer to voters always wins the election.

[15] Our focus on the class of monotonic equilibria is appealing because they are conceptually closer to simple rules of thumb that voters employ than more complicated non-monotonic strategies are. Since our model does not involve any strategic complementarity among voters' behavior, we cannot rule out mixed-strategy or non-monotonic pure-strategy equilibria. We believe that an analysis of such counter-intuitive equilibria will not contribute substantially to the conceptual development (or understanding) of manipulation in election campaigns.

Proposition 1 (Equilibrium without manipulation) *If manipulation is not feasible —i.e., $x_i = \theta + \varepsilon_i$— there is a unique electoral equilibrium. The voter behavior in this equilibrium is such that for any voter $i \in [0, 1]$, i votes for I if and only if her private signal is greater than or equal to $1/2$ —i.e., $s(x_i) = 0 \Leftrightarrow x_i \geq 1/2$. Moreover, whichever candidate stands closer to voters always wins the election.*

Proof See the beginning of the proof of Theorem 2.

When manipulative actions are allowed, we establish that there is still a unique monotonic equilibrium, where candidates engage in manipulative campaigns. Moreover, the manipulation level of the campaign of the candidate I is higher than that of candidate C. In the equilibrium, both the behavior of voters and the outcome of the election are determined by unique thresholds. Voters vote for the incumbent if and only if they observe a signal above a threshold x^*, and the incumbent wins the election if and only if the relative proximity parameter θ is above a threshold θ^*.

Theorem 1 (Equilibrium with manipulation) *Suppose that Assumption 1 holds. Given the state variable θ, there exists a unique monotonic electoral equilibrium in pure strategies. It is characterized by two thresholds x^* and θ^*, such that voter i supports candidate I if and only if her private signal is above x^*,i.e., $s(x_i) = 0 \Leftrightarrow x_i \geq x^*$, and I wins the election if and only if $\theta \geq \theta^*$ —i.e., I's political proximity to voters is sufficiently high relative to the proximity of candidate C. In this unique equilibrium, manipulative actions are strictly positive, and candidate I always manipulates to a higher extent than candidate C —i.e., $a_I(\theta) > a_C(\theta) > 0$ for any $\theta \in \mathbb{R}$— implying that net manipulation $\bar{a}(\theta)$ is strictly positive.*

Proof See the Appendix.

Theorem 1 establishes that there is a unique equilibrium in which (i) voters devise a monotonic strategy that takes into account the fact that the "war of disinformation" between candidates would distort the campaign signals in favor of the incumbent, and (ii) the candidates' best response to this common voting strategy and to their opponents' manipulative actions results in a net bias in campaign signals that favors the cost-efficient manipulator —i.e., the incumbent. That both candidates are compelled to do positive manipulation follows from the fact that for a given monotonic voting strategy, candidates can steal votes from each other by shifting the mean of the campaign signals. That the net manipulation $\bar{a}$ is strictly positive stems from I's cost

advantage, which enables her to distort the signals by more than what her opponent finds optimal. Aware of I's cost advantage, voters use a voting rule that tries to filter out the bias in campaign signals. This reinforces the necessity of manipulation from the candidates' point of view.

Theorem 1 by itself does not necessarily imply that voters are successfully manipulated. The natural question is whether the incumbent, through manipulation, can secure a victory that she would not be able to obtain in the absence of manipulation —i.e., a victory when $\theta < 1/2$. The theorem suggests that the answer depends on the magnitude of the unique threshold, θ^*, which, if sufficiently low —i.e., lower than 1/2— implies that the incumbent can get an undeserved victory whenever relative proximity of I satisfies $\theta \in [\theta^*, 1/2)$. The following theorem states the main result of the paper that manipulation can change the identity of the winner, whenever the cost parameters satisfy Assumption 1.

Theorem 2 *The threshold θ^* that determines the winner of the election is strictly smaller than the no-manipulation level of 1/2.*

Proof See the Appendix.

To illustrate how the threshold θ^* is determined, Figure 2 plots the numerical solutions to the equilibrium levels of net manipulation $\bar{a}(\theta)$ as a function of relative proximity of candidate positions to voters' bliss point.[16] We select the cost parameters and signal precision α to ensure that Assumption 1 is not violated. Therefore, the equilibrium is guaranteed to be unique and $\theta^* < 1/2$. The level of θ, at which the systematic bias $\bar{a}$ is at its maximum, coincides with θ^*.

When, for example, $\theta = 1/2$, so that the candidates are equally distant from voters, the latter would be indifferent between supporting I or C, had they been able to observe θ. Nonetheless, since $\theta^* < 1/2$, by Theorem 2, I receives a higher share of votes and wins the election. Only when θ is sufficiently low — i.e., below θ^*— does candidate C achieve a victory in the election; but even then, she gets a lower share of votes than she would obtain in the absence of manipulation.

[16] In all numerical solutions, we assume that voters and the candidates form their strategies as if the relative proximity θ of the incumbent candidate has a bounded support and can take a finite number of equally spaced values on the interval $[-1.5, 2.5]$ with a mid-point of 1/2. We also assume that the cost functions take the form $H_j(a) = \frac{1}{2}k_j a^2$ for $j = C, I$.

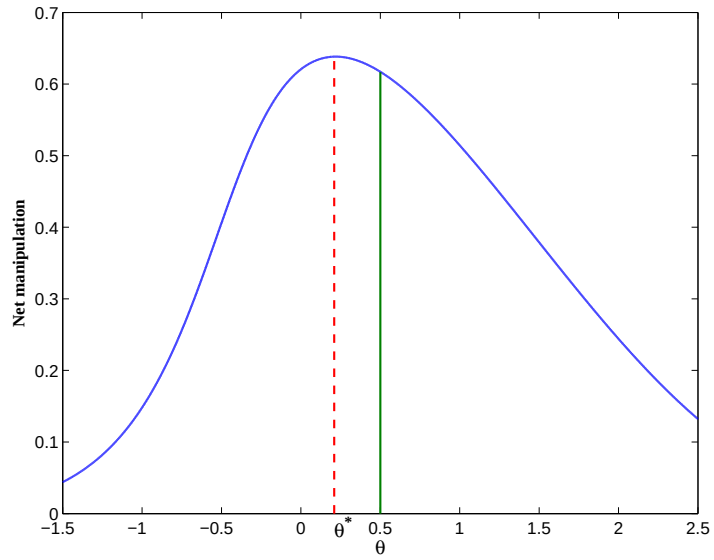

Figure 2: Equilibrium net manipulation ($\alpha = 1$, $k_I = 1$ and $k_C = 5$)

5 Why does manipulation work?

Comparing the electoral equilibria with and without manipulation, it is compelling to conclude that the only reason for the equilibrium bias in voters' beliefs is the presence of manipulative actions in an environment characterized by informational asymmetry between the candidates and the electorate. However, a closer look at the model will reveal that (i) the cost asymmetry between the candidates and (ii) an idiosyncratic noise in private signals under the lack of perfect communication among voters are equally critical for manipulation to be effective in equilibrium. In this section, we elaborate on why these conditions are crucial for effective manipulation, and conclude with a simple illustration of the inference problem they cause for the electorate.

5.1 Asymmetric manipulation costs

One of the assumptions that make manipulation effective is the asymmetry between manipulation costs of the candidates. Proposition 2 formalizes this point.

Proposition 2 *Suppose that the cost functions of the candidates are identical. Given the state variable θ, there exists a unique monotonic electoral equilibrium in pure strategies. It is characterized by two thresholds, x^* and θ^*, which are defined as in Theorem 1.*

In this unique equilibrium, manipulation actions, $a_I(\theta)$ and $a_C(\theta)$ are positive and equal to each other. Moreover, the thresholds x^* and θ^* are both equal to the no-manipulation level of $1/2$.

Proof The proof is omitted since it is embedded, as a special case, in the proofs of Theorem 1 and Theorem 2.

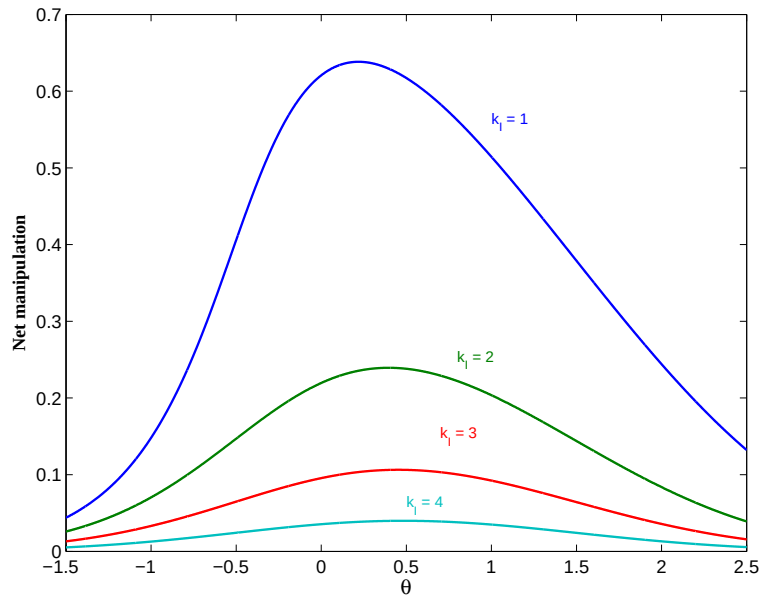

Figure 3: Equilibrium net manipulation as candidate I becomes more cost-efficient ($\alpha = 1$ and $k_C = 5$)

If the costs of I and C were same, then any manipulative attempt of either candidate would face an equivalent counter effect. Moreover, since no candidate would have a cost advantage, there would be no way to dominate the manipulation game. Thus, as stated in Proposition 2, net manipulation would be zero and thus voters can always identify the best candidate for them. Figure 3 demonstrates the uniform convergence of the net manipulation function to zero as the the cost difference between the candidates vanishes.

Note that the case of symmetric costs is different from the no-manipulation case. In this case, candidates do engage in manipulation to increase their vote share. Each candidate responds to the manipulative attempt of her opponent by more manipulation until the marginal benefit of manipulation equals its marginal cost. Thus, an

inefficiency arises with respect to the candidates' preferences. Both candidates would be better off if they were able to commit to zero manipulation instead of engaging in costly and wasteful manipulation.

5.2 Noisy campaign signals and imperfect communication

Without the uncertainty voters face about relative candidate proximity θ, there would clearly be no incentive for manipulative campaigns. What is less straightforward is the essential role of idiosyncratic noise in campaign signals in rendering manipulation effective and, hence, providing incentives for costly manipulation. In fact, voters' uncertainty about θ, by itself, would not be sufficient for the low-cost candidate to bias the posterior beliefs of voters in equilibrium. The consistency requirement of a perfect Bayesian equilibrium would rule out any such bias in the absence of some heterogeneity in private signals. To formally demonstrate this point, we next study the case of purely public signals, where each voter receives a common signal $x = \theta + a_I - a_C$ without any individual noise. Clearly, x becomes common knowledge after its realization.

The following proposition states that there is no equilibrium in which the cost-efficient candidate – i.e., candidate I – has an ex-ante advantage in the elections when voters receive a common public signal without noise (or idiosyncratic bias).

Proposition 3 *In the public-signal case, where $\varepsilon_i = 0$ for all $i \in [0, 1]$, there is no monotonic electoral equilibrium with manipulation.*

Proof See the Appendix.

Proposition 3 demonstrates that the results in Theorem 1 hinge on the assumption that voters observe manipulated signals with an idiosyncratic noise. A natural question then is how this individual noise in campaign signals enables the candidates to manipulate voters. When campaign messages contain an idiosyncratic noise, voters become uncertain about the population mean of the signals. Therefore, when evaluating their private signal x_i, they cannot perfectly differentiate between the systematic bias $\bar{a}$ and the individual bias ε_i that it contains. This informational problem gives the candidates an incentive to tailor their campaign messages in a particular way so as to bias voters' inference. In the next section, we discuss in more detail exactly how the cost-efficient candidate conceals part of the systematic bias $\bar{a}(\theta)$ in campaign signals behind a cloud of idiosyncratic noise and distorts voters' posterior beliefs in her favor.

5.3 Voters' inference problem

Recall that voters are uncertain about candidates' positions, while the candidates are able to observe θ and adjust their manipulation level based on its particular realization. When the communication environment that intermediates campaign messages adds an individual noise on manipulated public signals, voters cannot perfectly distinguish between the bias due to manipulative actions and the bias due to the individual noise that they cannot observe. Candidates exploit this additional uncertainty by tailoring their manipulative actions in such a way that voters' posterior beliefs remain biased in equilibrium. In particular, the shape of the equilibrium net manipulation function $\bar{a}(\theta)$ is such that voters end up overestimating the likelihood that θ is above $1/2$ when θ lies in a region where the incumbent's gain from manipulative actions is highest.

Figure 4 offers a simplified illustration of how this happens. The curve that lies above the 45-degree line describes how the public signal $\theta + \bar{a}(\theta)$ depends on the hidden state variable θ. It reflects the fact that the competition between incumbent and challenger candidates leads to a strictly positive net-bias function $\bar{a}(\theta) = a_I(\theta) + a_C(\theta)$. Although voters do not observe θ, since their beliefs cannot contradict what they know about the model, in equilibrium, they form their voting strategies based on the full knowledge of this function. As already illustrated through the numeric solutions depicted in Figure 2 and Figure 3, the shape of $\bar{a}(\theta)$ resembles a distorted normal curve. Therefore, the difference between public signals and the true value of the state variable θ increases as the latter approaches θ^* —i.e., the lowest level of θ under which the incumbent wins the election. Hence, the difference between the level of public signals and θ assumes its maximum value at θ^*. Beyond this point, $\bar{a}$ starts declining and so does the difference between public signals and θ.

If, as in Figure 4, the true state is $\theta_0 < 1/2$, then voters would be better off ex-post if the challenger had won the election. However, since $\theta_0 \geq \theta^*$, the incumbent ends up winning the election. To see why, consider two voters $i, j \in [0, 1]$. Under θ_0, the equilibrium level of public signal will be given by x_m. In the model, individual biases are normally distributed around a zero mean, and the normal curves in the figure represent the p.d.f.s of private signals x under different possible realizations of θ. For the sake of argument, suppose that the noise in signals can take one of two possible values, $\varepsilon > 0$ or $-\varepsilon < 0$, with equal probability. Voter i, who draws an individual bias ε, would only observe a signal $x_i = x_m + \varepsilon$. Note that her priors are such that we

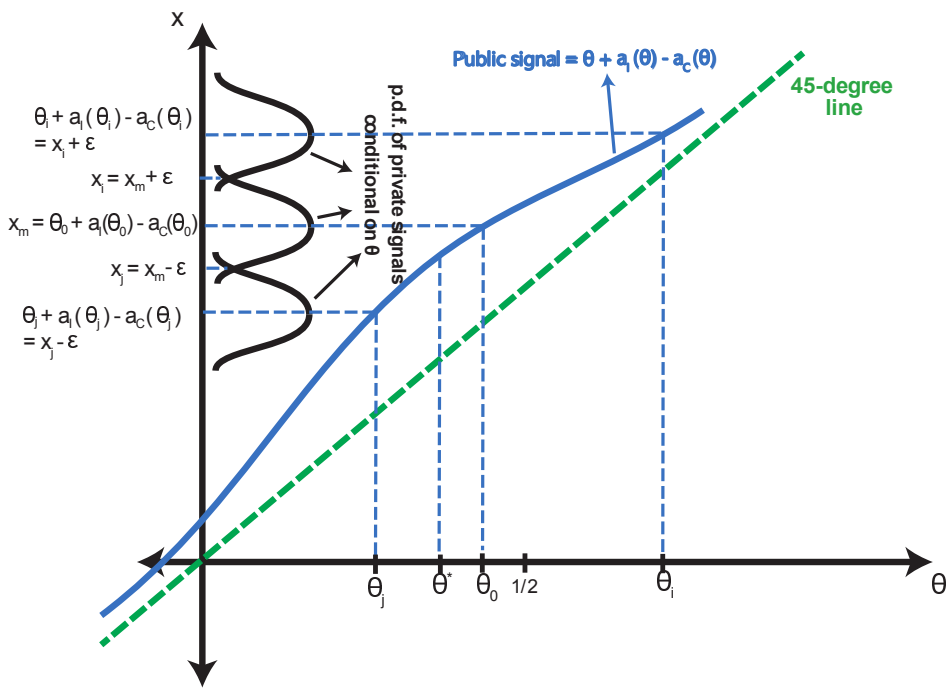

Figure 4: How does manipulation work?

can imagine that she assigns equal likelihood to all realizations of θ.[17] Knowing how equilibrium net manipulation behaves as a function of θ, she deduces that, conditional on observing x_i, the probability that she is in state θ_0 *and* has received a positive shock ε should be equal to the probability that the state is $\theta_i > \theta_0$ *and* has received a negative shock $-\varepsilon$. This equivalence is illustrated in Figure 4. Then, formally, we have $Pr(\theta = \theta_0|x_i) = Pr(\theta = \theta_i|x_i)$. Similarly, consider voter j, who receives $\varepsilon_j = -\varepsilon$. By the same logic, it must be that $Pr(\theta = \theta_0|x_j) = Pr(\theta = \theta_j|x_j)$. Therefore, for voter i [voter j], the event that $\theta = \theta_i$ [$\theta = \theta_j$] and $\theta = \theta_0$ will be assigned equal weights conditional on signal x_i [signal x_j].

Although voters i and j receive signals that are equally distant from x_m, due to the particular shape of the net manipulation function $\bar{a}(\theta)$, we have that $\theta_i - \theta_0 > \theta_0 - \theta_j$. Hence, if these voters were to form an expectation of θ conditional on their signals, voter i would overestimate θ by more than voter j would underestimate it. Obviously, in our model, a continuum of pairs of voters i and j will consider a continuum of possible combinations (θ, ε_i) and (θ, ε_j) that would result in signals x_i and x_j, respectively, when forming their expectations of the true state of the world. Yet, this would not

[17] Remember that in the model, voters were assumed to have uniform priors, whose support approaches to infinity, and which become uninformative at the limit.

change the conclusion that a larger mass of voters would end up believing that $\theta > 0.5$ and, therefore, voting for the incumbent when, in reality, $\theta = \theta_0 < 0.5$.

At this stage one might wonder why voters do not ignore the manipulative signals altogether. The answer lies in the fact that, in our model, manipulative campaign signals are the only sources of information for voters, and although these signals are somewhat misleading, they reveal some useful information about the hidden state variable θ. To clarify the point, consider a slightly modified model in which θ is priorly distributed according to normal distribution with mean 1/2. Then, if a voter ignores the signals, she will be left with her unbiased prior, which leaves her with a success rate of 1/2. However, if she takes the signal into account and forms her beliefs in the optimal way, then she expects to choose the wrong candidate only when θ falls into the interval $(\theta^*, 1/2)$. Yet, this occurs with a probability that is clearly smaller than 1/2.

6 Comparative statics and policy implications

In this section, we first numerically analyze how the equilibrium bias (net manipulation) in campaign messages, total election spending and the outcome of elections change in response to changes in the candidates' relative campaigning efficiency, as captured by the cost asymmetry in manipulation. To complement this analysis, we present a small extension of our main electoral model to demonstrate why limiting campaign contributions to political candidates may be desirable when costs of manipulation differ, and candidates must cater to non-median voter interests to access campaign funds. Then, we demonstrate through numerical simulations how a change in the degree of heterogeneity in private campaign signals (signal variance) affects the ability of the more-efficient candidate to manipulate the electorate over different regions of θ.

6.1 The degree of asymmetry in campaigning efficiencies

Gul and Pesendorfer (2012) show that, as the difference between candidates' costs of campaigning vanishes, voters' welfare increases. Theorem 1 had a similar implication. As the difference in marginal costs of campaigning diminishes, so does the level of net manipulation –as we have already seen in Figure 3–, and the systematic bias in voters' posterior beliefs disappears.

In most of the real settings, however, politicians and political parties are not equally efficient in spending their campaign resources. In our model, this fact has been cap-

tured by asymmetric marginal costs of manipulation between the candidates. This asymmetry can best be interpreted as a difference in campaigning efficiency between the candidates: Each additional dollar spent by the incumbent on manipulative actions leads to greater manipulation than a dollar spent by the challenger.

In the information age, rapid advancements in social communication technologies offer more-efficient campaigning tools, and candidates do not always catch up with these new innovations at the same rate. This may lead to a larger gap in campaigning efficiency. The 2008 U.S. presidential elections is a good example. In their book *The Obama Victory: How Media, Money, and Message Shaped the 2008 Election* Kate Kenski, Bruce W. Hardy, and Kathleen Hall Jamieson argue that among the main factors that gave the "Obama model" the upper hand against the McCain campaign were the former's relative effectiveness in the use of digital technology and the Internet to inform and mobilize the electorate and an innovative strategy of interpersonal, cable and radio micro-targeting to sway citizens that were more susceptible to persuasion.[18] In our model, such a widening of the efficiency gap between the candidates induces more manipulation by the advantaged candidate at any level of θ. Whether this has a direct negative impact on the identity of the election's winner -and, hence, on voters' welfare– depends on the true level of θ. If we denote the equilibrium winning thresholds before and after the increase in candidate I's manipulative efficiency by θ_0^* and θ_1^* respectively, then we have $\theta_1^* < \theta_0^* < 1/2$ because the net manipulation and the equilibrium bias increase in favor of candidate I following a reduction in k_I. Therefore voters will be adversely affected by an efficiency gain of candidate I if and only if $\theta \in [\theta_1^*, \theta_0^*]$. In other words, unless it is the the more-distant candidate who experiences the efficiency gain, and unless the decline in her marginal costs is sufficiently strong, the shift in voter shares will not be enough to make voters worse off.

However, this does not mean that voters are immune from indirect welfare costs associated with higher campaign spending. As evident from the left panel of Figure 5,

[18]The authors quote David Plouffe, the manager of Barack Obama's 2008 presidential campaign, recalling how, at some point, the campaign's e-mail list had reached 13 million people and how they directly reached about 20 percent of the votes they would need to win the elections by creating their own television network.The book also offers many interesting campaign-related statistics. For example, they cite a 2009 report by Edelman Research titled "The Social Pulpit: Obama's Social Media Toolkit," according to which Internet users spent over 14 million hours on YouTube watching more than 1,800 Obama campaign-related videos. The authors also report that during a typical general-election week, among the respondents to the National Annenberg Election Survey (NAES), those who reported having received campaign information from Internet sources in the previous week were 1.033 percentages more likely to vote for Obama.

the campaign costs of I rise over the entire range of θ as the cost parameter k_I decreases. This is because the convexity of the cost functions ensures that the increase in I's manipulative actions a_I more than offsets the reduction in costs due to higher efficiency.

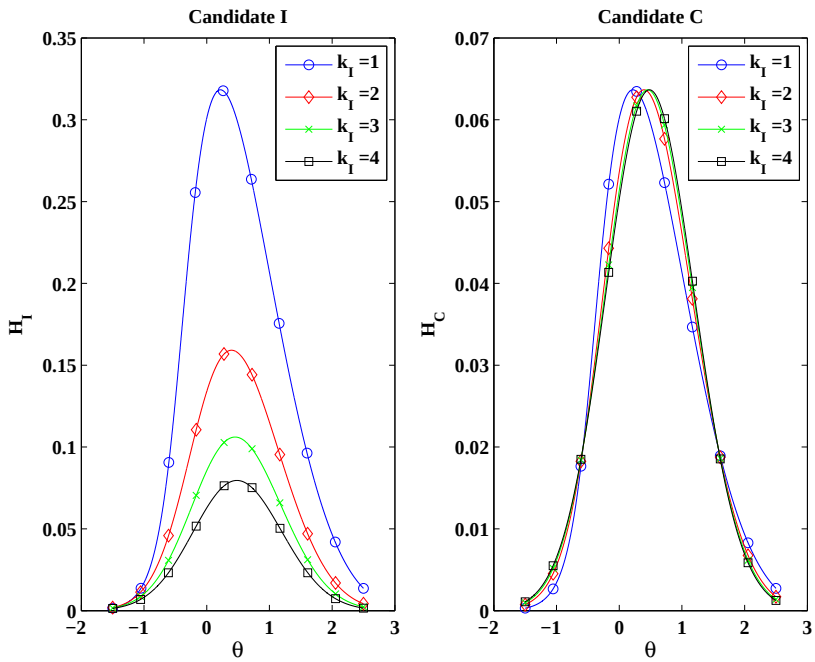

Figure 5: Campaign costs of the candidates as cost advantage of I increases ($\alpha = 1$, $k_C = 5$)

The less-efficient candidate, however, does not experience any change in her manipulative efficiency since k_C is assumed to stay constant. Therefore, the change in campaign costs of C is solely due to an adjustment of a_C as a response to the increase in a_I. As seen in the right panel of Figure 5, whether campaign costs of C rise or fall depends on the range of θ. When, I has some, but not too much, positional advantage over C in terms of the distance from voters, marginal gain for C from manipulative actions decreases as a_I goes up. Then, C chooses to lower a_C to economize on campaign costs. When, on the other hand, C has some positional advantage over I - i.e., when θ is sufficiently, but not too, low - the stakes for C in terms of potential votes she may lose to I is high. Therefore, a rise in her opponents' manipulation actions raises the marginal value of manipulation for C. At the extremes —i.e., when one candidate has substantial positional advantage over the other— qualitative conclusions are reversed. Since net manipulation is very low at these values of θ, only a very small mass of voters

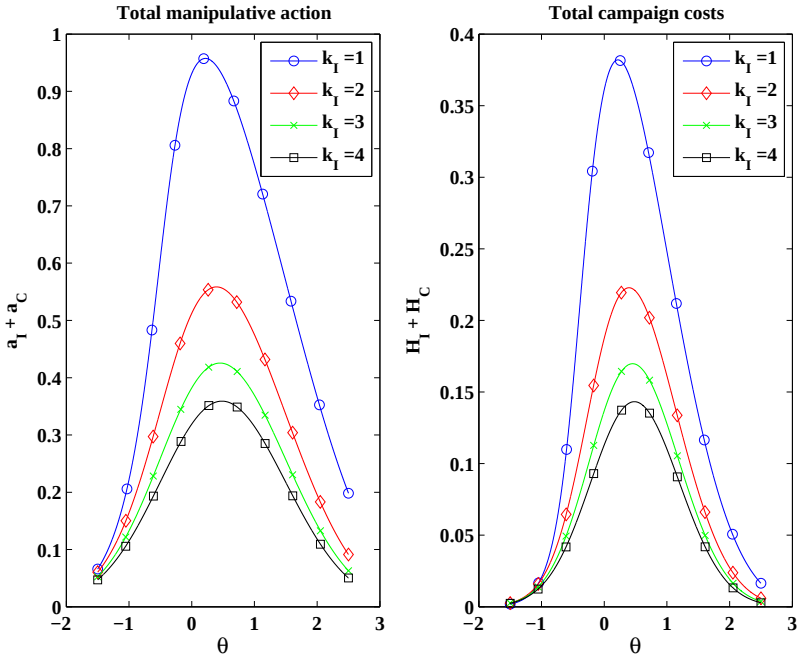

Figure 6: Total campaign costs and manipulative actions as cost advantage of I increases ($\alpha = 1$, $k_C = 5$)

—those at the tail of the normal distribution— receive signals that are near the voting threshold x^*. Therefore, the shift in voters' strategy x^*, triggered by the reduction in k_I, does not affect the marginal gain in votes from moving the mean of the campaign signals through manipulation as much as it does at intermediate values of θ.

Therefore, when she has substantial positional advantage, candidate C starts to spend less on manipulation because the increase in a_I is so minimal that reducing a_C leads to a fall in her costs that more than offsets her vote loss. However, when candidate I is significantly close to voters than C is, she raises a_I by enough for candidate C to feel compelled to respond with higher a_C.

Although candidate C's costs do not increase uniformly, Figure 6 suggests that total manipulation and total campaign spending by the two candidates unambiguously increase despite the fall in k_I. These numerical simulations suggest that a more pronounced incumbent advantage in the form of higher efficiency of campaign spending can be detrimental to social welfare both because wasteful campaign spending increase and because, in some circumstances, the incumbent may end up winning the election even when the other candidate was closer to voters' bliss point. The potential policy

implication is that reducing the cost asymmetry between the candidates can improve welfare not only by limiting disinformation caused by manipulation, but also by lowering the amount of total campaign spending.

6.2 Limits on campaign budgets

A typical model of informative political advertising finds that increasing costs of election campaigns harm voters since campaigns provide some information about candidates' characteristics or the state of the economy. Parallel to this simple intuition, Gul and Pesendorfer (2012) finds that if costs of campaigning increase for both candidates, voters' welfare decreases. They do not provide a welfare analysis for the case of distorted campaign signals, where candidates' actions may not provide any information. Similarly, Coate (2004a) shows that legal bounds on campaign spending transfer welfare from the median voter to interest groups because campaigns provide information to median voters. Prat (2002a, 2002b) study models of campaign finance with rational voters, politicians and lobbies, where political advertisements are informative only indirectly —i.e., campaign contributions by the lobbies serve as credible signals to voters about politicians' valence. In contrast to Coate (2004a) he concludes that a ban on campaign contributions may enhance the median voter's welfare under reasonable parameter values. Finally, Coate (2004b) proposes an alternative explanation for why limiting campaign contributions may lead to a Pareto improvement. His model highlights the trade-off candidates face between offering favors to interest groups to collect campaign contributions and the effectiveness of campaign spending in attracting swing voters. Campaigns are effective to the extent they induce swing voters, who are uninformed about candidate quality, to switch their votes from unadvertised candidate. When holding office is sufficiently valuable to the candidates, voters rationally believe that the former will implement more favors for their contributors hence undermining the effectiveness of campaigns. By lowering the level of spending, limits on campaign contributions can reduce the amount of favors promised to interest groups. As this would render campaigns more effective, the forced reduction in campaign spending does not necessarily lower the likelihood that a qualified candidate will be elected.

Our model departs from the previous studies in that we allow an electoral race between the candidates in which political advertising can be deceptive. Considering the comparative statics results in Figure 6 about the impact of cost asymmetry on net manipulation, one can easily infer that, under our setting, limiting campaign spending

can increase voter welfare by lowering net manipulation in campaign signals and reducing the bias in voter beliefs. In practice, a direct cap on manipulative actions would be infeasible, but a uniform upper bound on campaign spending, if binding for the incumbent candidate, would reduce the level of net manipulation and improve welfare. This is because a mandatory reduction in manipulation would be more pronounced for I than for C, and, therefore, net manipulation would decrease.[19]

To demonstrate that limiting campaign expenditures can be welfare improving also by promoting policy convergence towards voters' bliss point, we offer a simple extension of our original model in which access to campaign contributions are conditioned on the policy platform that candidates select. Suppose that voters are positioned at $\bar{\omega} = 0$, and each candidate $j \in P$ has to make a binary choice of political positions from $\{0, \omega_j\}$. We endogenize candidates' platforms in a rather reduced-form way. Assume that if any candidate j chooses ω_j instead of $\bar{\omega} = 0$, then she earns access to a budget of a fixed amount G, perhaps from special interest groups that are willing to fund the candidate if she positions herself at $\omega_j \neq 0$. We assume that G can be used for manipulation or be saved for later use, and this amount is given in the same units as the net benefit derived from vote shares. Suppose that $\omega_C < 0 < \omega_I$ are such that

$$\Phi(\sqrt{\alpha}(x^* - \theta_I - a_I(\theta_I))) < 1/2 < \Phi(\sqrt{\alpha}(x^* - \theta_C + a_C(\theta_C))), \tag{2}$$

where $\theta_I = 1/2 - |\omega_I|$, $\theta_C = 1/2 + |\omega_C|$ and x^* is the equilibrium voting strategy that voters would employ in the original model.[20] The first term θ_I is equal to the relative political position of candidates vis-à-vis voters when candidate C positions herself at $\omega_C = \bar{\omega} = 0$, while candidate I chooses $\omega_I > 0$. If, however, candidate C chooses the platform $\omega_C < 0$, while, this time, I positions herself at $\omega_I = \bar{\omega} = 0$, then $\theta = \theta_C$ instead. Equation (2) simply states that the political positions desired by special interest groups from each candidate –in exchange for access to campaign resources– are such that the candidates can raise their vote shares to above 1/2 if they unilaterally deviate from voters' bliss point and, at the same time, engage in

[19]Our model abstracts from potential benefits of election campaigns where some of the information provided to voters by the candidates is useful. Since regulators cannot perfectly distinguish between useful and manipulative information, there is inevitably a social trade-off between restricting manipulation and increasing the transmission of useful information.

[20]The assumption that voters ignore the state agency's behavior is not realistic. However, in our framework of imperfect social communication, even if voters take the state agency into account, they would not get perfectly precise signals about its possible behavior. Voters would use the signal about the state agency to further update their beliefs, but they would behave similarly anyway.

manipulation.

Suppose that there is a state agency that is entitled to limit the level of budget, G, to maximize voters' ex-post information about candidates' choice of political positions. In this case, the state agency will choose an upper-bound for G to guarantee that there is no manipulative action, assuming that voters will form their voting strategy as in the benchmark case —i.e., as if the manipulation game the candidates play is the same as in benchmark model. Formally:

Proposition 4 *On any equilibrium of the extended model, the state agency will set the amount of allowable campaign contributions G so that it satisfies*

$$G \leq -1 + 2\Phi(\sqrt{\alpha}(x^* - \theta_I - a_I(\theta_I))) + H_I(a_I(\theta_I))$$
$$G \leq 1 - 2\Phi(\sqrt{\alpha}(x^* - \theta_C + a_C(\theta_C))) + H_C(a_C(\theta_C)),$$

where x^*, $\{a_j(\theta_j)\}_{j \in P}$ are as defined in Theorem 1. Under this restriction, the candidates do not engage in manipulation.

Proof If either of the candidates $j \in P$ chooses ω_j, while the other chooses 0, which is the voters' position, then the condition in equation (2) guarantees that j wins the election. This lowers the voters' ex-post payoff compared to the case where both candidates choose 0. However, if the state agency chooses G such that the hypothesis holds, then no candidate would want to deviate from the case where both choose 0. Since bounding G as in the hypothesis is the only way to force candidates to coordinate on the no-manipulation equilibrium, the agency chooses an upper-bound on G as in the hypothesis. ■

When equation (2) holds, it means that if some candidate j deviates by choosing ω_j, then she will definitely win the election, although the other candidate stands closer to voters. Thus, the state agency wants to prevent such a deviation. If equation (2) does not hold, the agency will not care to satisfy the constraint on G given in Proposition 4. In other words, if, to access the resources, the candidates need to be so extreme that they would lose the election despite manipulative advertising, then there is no manipulation in the equilibrium as long as the winning candidate does not deviate from 0, too.

6.3 Increasing the precision of campaign signals

As already discussed in Section 5.2, the assumptions that manipulated signals arrive with a noise and that voters do not communicate with each other perfectly are crucial for manipulation to be effective. In this section, we discuss how a change in the degree of informational heterogeneity, as captured by the variance in private campaign signals, affects the electoral equilibrium.

A decrease in the variance of the idiosyncratic noise means that the campaign signals become more precise (higher α). One might be tempted to think that higher precision would lead to better information for voters by lowering the uncertainty about the real extent of the net bias (net manipulation) in campaign signals. However, this is not always the case. As we have already seen, the net manipulation declines as θ moves away from θ^* towards the extremes. Therefore, the mean campaign signal is closer to the voting threshold $x^* = \theta^* + \bar{a}(\theta^*)$ when θ is near θ^*. When campaign signals become more precise, relatively more voters will tend to receive signals that are closer to the mean signal $\bar{x}(\theta) = \theta + \bar{a}(\theta)$, which, if θ is near θ^*, means that a greater mass of signals will accumulate around the indifference threshold x^*. Therefore, a given shift in the mean campaign signal through higher manipulation allows the candidates to steal a larger fraction of votes than before.

As θ moves towards the extremes, at some point, the mean signal will be sufficiently distant from x^* so that only those voters at one of the far tails of the signal distribution will be indifferent between voting for I and C. Moreover, their size will shrink further with greater signal precision, which, in turn, will lower the marginal gain in vote share from manipulative actions and hence lead to lower net manipulation.

To illustrate the impact of increased signal precision on election outcomes, Figure 7 plots the share of votes the less-efficient candidate (candidate C) receives under different combinations of k_I and α values, when both candidates are situated in equal proximity to voters' bliss point (i.e., when $\theta = 1/2$). As before, we let k_I take four different values, while α increases from 0.5 to 2 in three equal increments.[21] As seen in the figure, as long as k_I remains below k_C, the incumbent receives a strictly higher share of the votes, although candidates are equally distant from the common voter position. Challenger's vote share falls further when the cost-asymmetry increases. At the current value of θ, a rise in precision leads to higher net manipulation and lower vote

[21] Given the combinations of k_I and k_C that we consider here, α should remain below a threshold that is approximately equal to 2.06. Otherwise, Assumption 1 would be violated.

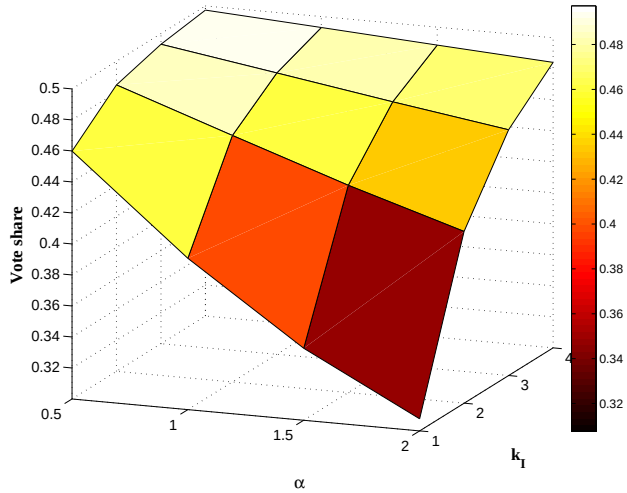

Figure 7: Vote share of C in equilibrium when $\theta = 0.5$ ($k_C = 5$)

share for C. At least over a certain range of signal variances, campaign signals that are more spread out across the population (have higher variance) tend to discourage manipulation of voters. From a broad policy perspective, this is a novel explanation for how a more decentralized media might be welfare-improving.

7 Extensions

7.1 Strategic Voting

In the benchmark model, we have assumed that voters choose a candidate independently of how others vote. This assumption is natural when there is a continuum of voters, and hence the measure of each voter is ignorable. However, when the population is finite, rational voters will know that their votes matter only if they are pivotal. In that case, it is a weakly dominant strategy for them to condition their choice on being the pivotal voter. For a voter to be pivotal, exactly half of the remaining voters should have received signals favoring one candidate while the other half must have received messages that favor the other candidate. We assume that the size of the electorate is given by a positive integer N, which equals to $2l+1$, where l is a positive integer. Note that if there were an even number of voters, being pivotal would be a zero-probability event, and we would be back in our benchmark model without strategic voting. Assuming that the electorate uses monotonic strategies, a voter makes her decision as if

exactly l voters have received a signal greater than the expected signal threshold $\hat{x}$, and the remaining l voters have received a signal smaller than $\hat{x}$. Then, having observed her private signal x_i, the voter's posterior belief will no longer be

$$P(\theta|x_i, \hat{x}) = \frac{\phi(\sqrt{\alpha}(x_i - \bar{a}(\theta) - \theta))d\theta}{\int_{-\infty}^{\infty} \phi(\sqrt{\alpha}(x_i - \bar{a}(\theta) - \theta))d\theta},$$

but instead, it will be equal to

$$\widetilde{P}(\theta|x_i, \hat{x}) = \frac{\phi(\sqrt{\alpha}(x_i - \bar{a}(\theta) - \theta))\Phi(\hat{\nu})^l(1 - \Phi(\hat{\nu}))^l d\theta}{\int_{-\infty}^{\infty} \phi(\sqrt{\alpha}(x_i - \bar{a}(\theta) - \theta))\Phi(\hat{\nu})^l(1 - \Phi(\hat{\nu}))^l d\theta}, \tag{3}$$

where $\Phi(\cdot)$ is the c.d.f. of normal distribution and $\hat{\nu} = \sqrt{\alpha}(\hat{x} - \bar{a}(\theta) - \theta)$. Note that the main difference between the two posterior functions is an additional curvature introduced by the term $\Phi(\hat{\nu})^l(1 - \Phi(\hat{\nu}))^l$ in the pivotal voter's posterior belief function $\widetilde{P}(.|.,.)$. This additional curvature is the result of the additional information that the voter incorporates in her posterior belief.

It is straightforward to show that, except this modification to voters' posterior beliefs, nothing else is different between the strategic voting framework and our benchmark model with a continuum of voters. Candidates' behavior remains qualitatively unchanged as long as they expect that voters' strategies are monotonic. As a result, we can state that there exists a monotonic equilibrium, in which the net manipulation and the systematic bias are both positive. However, the method we used to prove the uniqueness of the equilibrium in the benchmark model does not directly apply to this new setting, because now the voters' posteriors take a different form.

Posterior beliefs, which are formed under the assumption of being pivotal, incorporate new information regarding the position of other voters' private signals. This is reflected in flattening out of the posterior density functions and results in a higher equilibrium threshold x^* for idiosyncratic signals. Strategic voting implies a setting where the information sets, on which voters condition their behavior, overlap to a greater extent, and this leads to greater coordination among voters.

Since, the net manipulation function remains qualitatively unchanged, the increase in the equilibrium voting threshold x^* reduces the systematic bias, $1/2 - \theta^*$. Indeed, as the number of voters increases, the quality of the information that each voter obtains is improved. This in turn leads to lower systematic bias in beliefs in favor of the cost-efficient candidate. Figure 8 shows the relationship between the number of voters and the systematic bias. Note that when there is only one voter, being pivotal clearly does

not provide any additional information. Thus, the systematic bias exactly equals the bias in the infinite voters case.

On the other hand, when there are multiple voters that communicate with each other and share the private signals they receive, voters' information about the mean signal becomes more precise. Under the extreme scenario of perfect communication, voters effectively observe the private signals received by each and every other voter. Using this information, they obtain a more precise information about the hidden state θ. Since in such a case, all voters will have the same information, there will be no heterogeneity among voters and they will behave collectively like a single voter. In that case, being pivotal becomes a zero-probability event and thus we are back to the benchmark case, except that the precision of private signals increases as the number of voters increases. Therefore, the systematic bias increases as well.

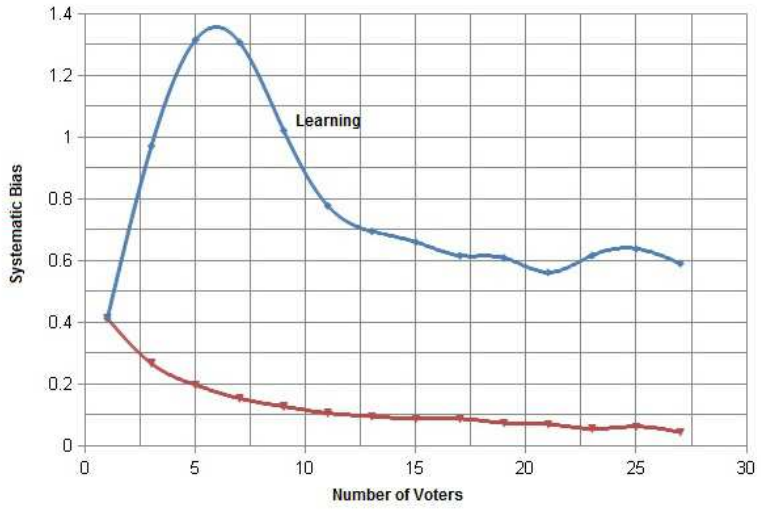

Figure 8: Impact of the number of voters on Systematic Bias

Now suppose that search and communication costs prevent perfect communication among voters. Then, some degree of informational heterogeneity among voters will remain, since the sample of signals they collect will not fully coincide. This way, being pivotal will continue to be a probable event, and the precision of the posterior beliefs of voters will increase with the number of voters. In this scenario, increasing precision leads to more effective manipulation. However, at the same time, the quality of the information associated with being pivotal also increases. Thus, increasing the number of voters will result in two opposing effects on the systematic bias in posterior beliefs.

Accordingly, as illustrated by the curve labeled as "Learning" in Figure 8, when the size of the electorate gets larger, the systematic bias initially increases and then it starts to decrease and approaches a steady level.

7.2 Heterogenous Preferences

In the benchmark model, we have assumed that the voters are heterogeneous with respect to their private information, but not in their political preferences regarding ideal candidate positions. Each voter had her own particular position in the social communication environment and received a different signal about the quality of the candidates. In this section, we allow for preference heterogeneity among voters in addition to informational heterogeneity.

Now, suppose that there are a continuum of independently and identically distributed random variables indexed by the set of voters. For each $i \in [0,1]$, $\rho_i \sim \mathcal{N}(\frac{1}{2}, \frac{1}{\beta})$, and β characterizes the concentration of preferences among voters. For each voter i, ρ_i denotes the threshold over which candidate I becomes a better alternative than C for voter i. That is, i prefers I over C if and only if

$$P(\theta > \rho_i | x_i) > 1/2,$$

where, as before, x_i is the private signal that i receives. Then the vote share of I is

$$|\{i \in [0,1] : P(\theta > \rho | x_i) > 1/2\}| = Prob\left((x,\rho) : P(\theta > \rho | x) > 1/2\right),$$

where the equality follows from the Law of Large Numbers. Since the preferences and private signals are independent, we can write this share as follows:

$$\int_{-\infty}^{\infty} \int_{\{x:P(\theta>\rho|x)>1/2\}} \sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho - \frac{1}{2}\right)\right)\sqrt{\alpha}\phi(\sqrt{\alpha}(x - \bar{a}(\theta) - \theta))dxd\rho. \quad (4)$$

Assuming that every voter chooses a monotonic strategy, each voter, i, will have a threshold $\hat{x}_i$ for her voting decision. Thus, the preference parameters induce a set of independently and identically distributed random variables indexed by the set of voters, $\{\hat{x}_i\}_{i\in[0,1]}$, such that

$$P(\theta > \rho_i | x_i) > 1/2 \Leftrightarrow x_i > \hat{x}_i = \hat{x}(\rho_i).$$

In that case, we can rewrite equation (4) as

$$\int_{-\infty}^{\infty} \sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho - \frac{1}{2}\right)\right)(1 - \Phi(\sqrt{\alpha}(\hat{x}(\rho) - \bar{a}(\theta) - \theta)))d\rho. \tag{5}$$

Then the first order condition for candidate I's vote share maximization problem will be

$$2\int_{-\infty}^{\infty} \sqrt{\beta}\sqrt{\alpha}\phi\left(\sqrt{\beta}\left(\rho - \frac{1}{2}\right)\right)\phi(\sqrt{\alpha}(\hat{x}(\rho) - \bar{a}(\theta) - \theta))d\rho = k_I a_I(\theta). \tag{6}$$

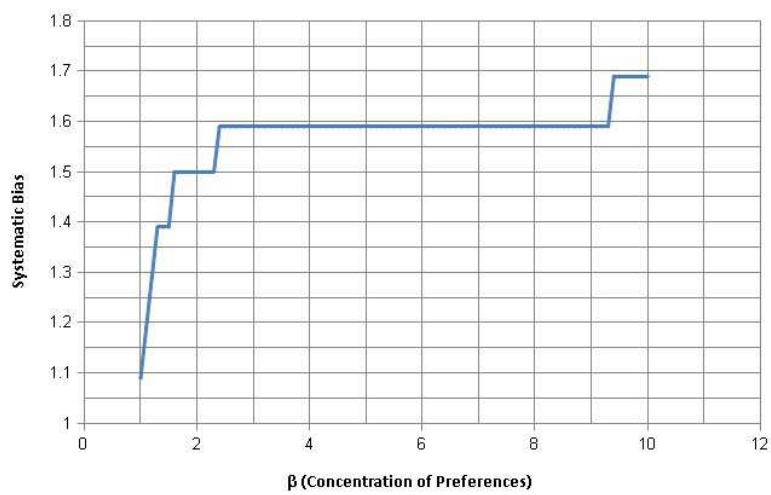

Figure 9: Concentration of preferences increases systematic bias

Following the steps of the proof of Theorem 1, we can prove a parallel result for heterogeneous voters as well:

Theorem 3 *Suppose that Assumption 1 holds. Given the state variable θ, there exists a unique monotonic electoral equilibrium in pure strategies. It is characterized by the set of thresholds $\{\hat{x}_i\}_{i\in[0,1]}$. In this unique equilibrium, manipulative actions are strictly positive, and candidate I always manipulates to a higher extent than candidate C — i.e., $a_I(\theta) > a_C(\theta) > 0$ for any $\theta \in \mathbb{R}$— implying that the net manipulation $\bar{a}(\theta)$ is strictly positive.*

Proof See the Appendix.

Figure 9 shows how an increase in the concentration of preferences around a given political bliss point affects the impact of manipulation on the systematic bias in electorate's voting behavior. As the distribution of preferences of voters becomes more concentrated, systematic bias due to manipulation increases. As the source of preference heterogeneity among voters becomes weaker, the strategic advantage of the cost-efficient candidate (candidate I) increases.

8 Conclusion

This paper has proposed a theoretical model of electoral competition between two candidates who have incentives to send costly manipulative signals to bias public opinion and increase their voter shares. Due to the information asymmetry between the electorate and the candidates, voters have to base their decisions on campaign signals about candidate positions. These positions are both systematically manipulated and contain a private noise due to the heterogeneous relationship of voters with the social communication structure that intermediates political messages. In other words, voters face two uncertainties: They are uncertain about both the candidates' real political platforms and the location of their private signal in population distribution of signals. These two factors, combined, make voters unable to filter out the bias correctly and create an incentive for politicians to send manipulated signals. Given such an incentive, the difference in the candidates' cost structures (campaigning efficiency) results in a positive net bias in voters' posterior beliefs in favor of the more-efficient candidate. Under such a setting, we prove that monotonic equilibria involve positive manipulation by both candidates, and the candidate with a cost advantage is capable of biasing the electorate's posterior beliefs and influencing the election outcomes in her favor.

We have taken a simplistic approach when modeling political propaganda by assuming that all of the candidates' announcements and actions, which, in reality, refer to multiple political dimensions, are aggregated into signals that can be interpreted as claims of political closeness to voters. This allowed us to model the competition between the candidates as a race to shift the value of a single indicator. We have also abstracted from messages that do not aim at manipulating the voters' opinion, but contains truthful messages. Although we have discussed the welfare implications of our model, to address questions about optimal campaign regulations in the context of a formal welfare analysis, it is necessary to explicitly model social costs of campaign

spending, as well as the benefits and losses associated with election outcomes. An interesting extension left for future work is to combine informative and manipulative political campaigns in a single model.

Appendix: Proofs

Proof of Theorem 1 The proof consists of three stages. In the first stage, we show that, given monotonic strategies of voters, there is a unique net manipulation function which is necessarily a positive, bounded and continuously differentiable function of θ. In the second stage, we show that, given any positive, bounded and continuously differentiable net manipulation function, voters uniformly use a monotonic strategy as a function of private signals. In the third stage, we will see that what we have in the first two stages requires a unique equilibrium.

Stage 1: Given a monotonic voting strategy $\hat{x}$ of the electorate, the set of idiosyncratic random noises, ε, that will induce a voter to vote for candidate C is

$$\{\varepsilon | \varepsilon < \hat{x} - \theta - a_I + a_C\}.$$

The size of supporters for C can be written as follows, by the Law of Large Numbers,

$$S(a_I, a_C; \theta) = \Phi(\sqrt{\alpha}(\hat{x} - \theta - a_I + a_C)) \tag{7}$$

Given the challenger's action a_C and θ, the net return to manipulation for candidate I is

$$u_I = (a, a_C; \theta) = 1 - 2\Phi(\sqrt{\alpha}(\hat{x} - \theta - a_I + a_C)) - H_I(a).$$

From now on we will assume that for each candidate l $H_l(x) = \frac{1}{2}\ k_l x^2$, where the cost parameters satisfy Assumption 1.

When manipulation level is zero, the return to manipulative action is

$$2\sqrt{\alpha}\phi(\sqrt{\alpha}(\hat{x} - \theta + a_C));$$

however, the cost of manipulative action is zero for for zero manipulation level. Thus there is $a_I > 0$ such that net return of a_I is greater than net cost of a_I, which implies that there is no corner solution for candidate I's vote maximization problem. A similar

argument shows that there is no corner solution to challenger's problem as well.

Then the first order conditions are necessary for solution. This implies, for each θ, that

$$k_C a_C(\theta) = 2\sqrt{\alpha}\phi(\sqrt{\alpha}(\hat{x} - a_I(\theta) + a_C(\theta) - \theta)) = k_I a_I(\theta). \tag{8}$$

The net manipulation function $\bar{a}(\theta) = a_I(\theta) - a_C(\theta)$ is given as follows:

$$\bar{a}(\theta) = \frac{k_C - k_I}{k_I k_C}\sqrt{\frac{2\alpha}{\pi}} e^{-\frac{1}{2}\alpha(\hat{x} - \bar{a}(\theta) - \theta)^2} \tag{9}$$

Let $\hat{\nu}$ denote $\sqrt{\alpha}(\ \hat{x} - \bar{a}(\theta) - \theta)$ and

$$A = \frac{k_C - k_I}{k_I k_C}\sqrt{\frac{2\alpha}{\pi}}.$$

The right-hand side of equation (9) is positive, for any net manipulation level $\bar{a} \geq 0$ and uniformly bounded. The left-hand side equals 0 when $\bar{a} = 0$ and diverges linearly to infinity as $\bar{a}$ diverges to infinity as well. This implies that there always exists a positive solution to equation (9). Then by Axiom of Choice, from set theory, there exists a well-defined, positive and bounded net manipulation function over the set of relative policy positions of candidates.

To ensure the uniqueness and smoothness of the net manipulation function we will need some further restrictions on the cost parameters.

The derivative of the function

$$a - Ae^{-\frac{1}{2}\hat{\nu}^2},$$

with respect to a is positive when

$$\sqrt{\alpha} A \hat{\nu} e^{-\frac{1}{2}\hat{\nu}^2} < 1. \tag{10}$$

Since

$$\hat{\nu} e^{-\frac{1}{2}\hat{\nu}^2} < \frac{1}{\sqrt{e}},$$

condition in equation (10) amounts to

$$k_I \geq \left(\sqrt{\frac{2}{e\pi}}\right)\alpha,$$

which holds by Assumption 1.

Then for every θ there exists a unique, positive net manipulation level that satisfies equation (9). By implicit function theorem, the net manipulation function is than continuously differentiable.

Before passing to stage 2, we will establish that the function $y(\theta) := \bar{a}(\theta) + \theta$ is invertible. Since the net manipulation function $\bar{a}(\cdot)$ is differentiable, $y(\cdot)$ is differentiable as well. Then for invertibility of $y(\cdot)$, it is sufficient to show that the derivative of $y(\cdot)$ is always positive, which is equivalent to condition that $\bar{a}'(\theta) > -1$ for any θ.

The derivative of the net manipulation function is given by

$$\bar{a}'(\theta) = \frac{\sqrt{\alpha}A\hat{\nu}e^{-\frac{1}{2}\hat{\nu}^2}}{1-\sqrt{\alpha}A\hat{\nu}e^{-\frac{1}{2}\hat{\nu}^2}}, \tag{11}$$

which is always greater than -1 by Assumption 1.

Manipulating equation (11) gives for any θ that

$$\bar{a}'(\theta) \in \left[-\frac{\sqrt{\alpha}A}{\sqrt{e}+\sqrt{\alpha}A}, \frac{\sqrt{\alpha}A}{\sqrt{e}+\sqrt{\alpha}A}\right],$$

thus

$$y'(\theta) \in \left[\frac{\sqrt{e}}{\sqrt{e}+\sqrt{\alpha}A}, \frac{\sqrt{e}}{\sqrt{e}-\sqrt{\alpha}A}\right]. \tag{12}$$

Second stage:

Now we assume that voters expect to face a net manipulation function, which is continuously differentiable, bounded and positive. The posterior probability that $\theta \geq \frac{1}{2}$ as a function of the private signal, x_i, that each voter, i, receives is

$$P(x_i) := P(\theta \geq \frac{1}{2}|x_i) = \frac{\int_{\frac{1}{2}}^{\infty}\sqrt{\alpha}\phi(\sqrt{\alpha}(x_i-\bar{a}(\theta)-\theta))d\theta}{\int_{-\infty}^{\infty}\sqrt{\alpha}\phi(\sqrt{\alpha}(x_i-\bar{a}(\theta)-\theta))d\theta} \tag{13}$$

The derivative of $P(\cdot)$ is positive if and only if

$$\int_{-\infty}^{\frac{1}{2}}\nu_i\phi(\nu_i)d\theta\int_{\frac{1}{2}}^{\infty}\phi(\nu_i)d\theta < \int_{\frac{1}{2}}^{\infty}\nu_i\phi(\nu_i)d\theta\int_{-\infty}^{\frac{1}{2}}\phi(\nu_i)d\theta.$$

When we apply change of variables to $\int \nu_i \phi(\nu_i) d\theta$, we get that

$$\int_{-\infty}^{\frac{1}{2}} \nu_i \phi(\nu_i) d\theta = -\int_{-\infty}^{\sqrt{\alpha}(y(\frac{1}{2})-x_i)} \frac{-\nu_i \phi(\nu_i) d\nu_i}{y'(\theta)} < \\ -\frac{\phi(\sqrt{\alpha}(y(\frac{1}{2})-x_i)(\sqrt{e}-\sqrt{\alpha}A)}{\sqrt{e}} < 0, \tag{14}$$

where the last inequality follows from equation (10).

Similarly

$$\int_{\frac{1}{2}}^{\infty} \nu_i \phi(\nu_i) d\theta = -\int_{\sqrt{\alpha}(y(\frac{1}{2})-x_i)}^{\infty} \frac{-\nu_i \phi(\nu_i) d\nu_i}{y'(\theta)} < \\ \frac{\phi(\sqrt{\alpha}(y(\frac{1}{2})-x_i)(\sqrt{e}-\sqrt{\alpha}A)}{\sqrt{e}} > 0, \tag{15}$$

Since $\int \phi(\nu_i) d\theta > 0$, equations (14) and (15) jointly show that $P'(x_i) > 0$ for any x_i.

Now $x_i \to \infty$ implies that the segment of the function $\phi(\sqrt{\alpha}\ (x_i - \bar{a}(\theta) - \theta))$ converges uniformly to zero, which implies the convergence in probability as well. Thus

$$Lim_{x_i \to \infty} P(x_i) = 1.$$

Similarly $x_i \to -\infty$ implies that $P(x_i)$ converges to 0. Thus by intermediate value theorem and the monotonicity of $P(\cdot)$, there exists a unique $\hat{x}$ that satisfies $P(\hat{x}) = 1/2$.

Stage 3: Combining the results in stage 1 and stage 2, we can conclude that there must be a unique equilibrium $(x^*, \bar{a}(\theta; x^*))$ where the equilibrium voting strategy x^* is a best response to equilibrium net manipulation function $\bar{a}(\theta; x^*) > 0$ and vice versa. Since $y(\theta)$ is strictly increasing, there must also be unique threshold for relative candidate proximity θ^* under which both candidates obtain an equal share of votes, i.e., $\Phi(\sqrt{\alpha}(x^* - \theta^* - \bar{a}(\theta^*; x^*))) = 1/2$. This completes the proof. ■

Proof of Theorem 2 When, manipulation is not feasible, it is straightforward to show that the posteriors of voters can be represented by c.d.f. of normal distribution. Since $\Phi(\cdot)$ is a strictly increasing and continuous function, by intermediate-value theorem, there is a unique x^* such that $\Phi(\sqrt{\alpha}(x_i - 1/2)) \leq 1/2$ if and only if $x_i \leq x^*$.

Since $\Phi(0) = 1/2$, $x^* = 1/2$. Then, the vote share of C is $S(\theta) = \int_{-\infty}^{1/2} f(x_i|\theta)dx_i = \int_{-\infty}^{1/2} \sqrt{\alpha}\phi(\sqrt{\alpha}(x_i - \theta))dx_i = \Phi(\sqrt{\alpha}(1/2 - \theta))$. Then, $S(\theta) > 1/2$ if and only if $\theta < 1/2$, as required.

Whenever, the manipulation is allowed, the threshold θ^* that determines the winner of the election is the solution to the following equation

$$\Phi(\sqrt{\alpha}(x^* - \bar{a}(\theta^*) - \theta^*)) = \frac{1}{2}, \tag{16}$$

where $\Phi(\cdot)$ is the cumulative normal distribution function. Note that equation (16) is satisfied if and only if

$$x^* - \bar{a}(\theta^*) - \theta^* = 0. \tag{17}$$

Then,

$$\bar{a}(\theta^*) = Ae^{-\frac{1}{2}\alpha(x^* - \theta^* - \bar{a}(\theta^*))^2}, \tag{18}$$

which implies that

$$\bar{a}(\theta^*) = A. \tag{19}$$

Note that A is the maximum value that the net manipulation may ever take. Actually, inspecting equation (11) reveals that θ^* is the only maximizer of the net manipulation function. Moreover, it is straightforward to show that $\theta < \theta^* \Rightarrow \bar{a}'(\theta) > 0$ and $\theta > \theta^* \Rightarrow \bar{a}'(\theta) < 0$.

Now, in equilibrium

$$\frac{\int_{\frac{1}{2}}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta}{\int_{-\infty}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta} = \frac{1}{2},$$

thus

$$\int_{\frac{1}{2}}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta = \int_{-\infty}^{\frac{1}{2}} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta. \tag{20}$$

Let $\nu^\theta = \sqrt{\alpha}(\ \theta + \bar{a}(\theta) - x^*)$, and $\nu^* = \sqrt{\alpha}(\ \theta^* + \bar{a}(\theta^*) - x^*) = 0$. Then

$$\frac{d\nu^\theta}{d\theta} = \sqrt{\alpha}(1 + \bar{a}'(\theta)) > 0,$$

and when $\theta = 1/2$

$$\nu^{\frac{1}{2}} = \sqrt{\alpha}\left(\frac{1}{2} + \bar{a}(1/2) - x^*\right).$$

Now, suppose that $\theta^* \geq \frac{1}{2}$. Then, $\nu^* \geq \nu^{\frac{1}{2}}$. Thus

$$\int_{\frac{1}{2}}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta =$$
$$= \int_{\frac{1}{2}}^{\theta^*} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta + \int_{\theta^*}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta,$$

where the first term is non-negative and the second term can be expressed as follows by means of change of variables:

$$\int_{\nu^*}^{\infty} \frac{\phi(\nu^\theta)d\nu^\theta}{(1 + \bar{a}'(\theta))} = \int_{0}^{\infty} \frac{\phi(\nu^\theta)d\nu^\theta}{(1 + \bar{a}'(\theta))}.$$

Since $\nu^\theta \geq \nu^*$ if and only if $\theta \geq \theta^*$, we conclude that $\bar{a}'(\theta) < 0$ in the integral above. Thus,

$$\int_{0}^{\infty} \frac{\phi(\nu^\theta)d\nu^\theta}{(1 + \bar{a}'(\theta))} > \int_{0}^{\infty} \phi(\nu^\theta)d\nu^\theta = \frac{1}{2}.$$

This result implies that

$$\int_{\theta^*}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta > \frac{1}{2},$$

and hence,

$$\int_{\frac{1}{2}}^{\infty} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta > \frac{1}{2}.$$

On the other hand, a symmetric argument shows that

$$\int_{-\infty}^{\frac{1}{2}} \sqrt{\alpha}\phi(\sqrt{\alpha}(x^* - \theta - \bar{a}(\theta)))d\theta < \frac{1}{2},$$

which contradicts with equation (20). This contradiction implies that $\theta^* < 1/2$.

Proof of Theorem 3 The share of votes for candidate I is given by the following equation

$$\int_{-\infty}^{\infty}\int_{x:P(\theta>\rho|x)>1/2}\sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho-\frac{1}{2}\right)\right)\sqrt{\alpha}\phi(\sqrt{\alpha}(x-\bar{a}(\theta)-\theta))dxd\rho \tag{21}$$

Assuming that each voter plays a monotonic strategy $\hat{x}_i = \hat{x}(\rho)$, then we can write equation (21) as

$$\int_{-\infty}^{\infty}\sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho-\frac{1}{2}\right)\right)\left[1-\Phi(\sqrt{\alpha}(x-a_I(\theta)+a_C(\theta)-\theta))\right]d\rho \tag{22}$$

Then the first order condition for candidate I's vote share maximization problem will be then:

$$2\int_{-\infty}^{\infty}\sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho-\frac{1}{2}\right)\right)\sqrt{\alpha}\phi(\sqrt{\alpha}(x-a_I(\theta)+a_C(\theta)-\theta)) = k_I a_I(\theta) \tag{23}$$

The first order condition for candidate C's problem can be determined similarly, which implies that net manipulation function is the solution to the following implicit equation:

$$\bar{a}(\theta) = \frac{k_C-k_I}{k_C k_I}2\int_{-\infty}^{\infty}\sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho-\frac{1}{2}\right)\right)\sqrt{\alpha}\phi(\sqrt{\alpha}(x-\bar{a}(\theta)-\theta))d\rho \tag{24}$$

The right-hand side of equation (24) is always positive and bounded; however, the left-hand size equals zero when the net manipulation level is zero and diverges linearly to infinity as the net manipulation increases indefinitely. Thus, there is always a positive solution to equation (24). Moreover, the derivative of the right-hand side of equation (24) is

$$\bar{a}(\theta) = \frac{k_C-k_I}{k_C k_I}2\int_{-\infty}^{\infty}\sqrt{\beta}\phi\left(\sqrt{\beta}\left(\rho-\frac{1}{2}\right)\right)\alpha(x-\bar{a}(\theta)-\theta)\phi(\sqrt{\alpha}(x-\bar{a}(\theta)-\theta))d\rho, \tag{25}$$

which is less than unity by Assumption 1. Thus for each θ and the set of thresholds $\{\hat{x}(\rho)\}_{\rho\in\mathbb{R}}$, there is a unique solution to equation (24). This implies that net manipulation function is well-defined, bounded, and continuously differentiable. Moreover, equation (25) implies that the derivative of $\bar{a}(\theta)$ is greater than -1, which requires that

$y(\theta) = \bar{a}(\theta) + \theta$ is invertible for any θ.

The second and third stages of the proof follows same steps as the proof of Theorem 1.

Proof of Proposition 3 We will consider the pure-strategy and mixed strategy cases separately.

Pure-Strategy Equilibria

When $\varepsilon_i = 0$ for any voter i, all voters receive the same signal, which unifies their behavior. Thus the possible vote shares are only the degenerate ones: either all voters support I or none of them. Then, candidates engage in costly manipulation only if they expect to change the election outcome. The benefit of winning the election for the candidates is 2. Thus, candidates will find it preferable to manipulate until the manipulation cost is less than 2.

Now suppose that voters adopt a monotonic strategy with the threshold of $\hat{x}$. Then, for any θ smaller than $\hat{x} - \frac{2}{\sqrt{k_I}}$ and greater than $\hat{x} + \frac{2}{\sqrt{k_C}}$, neither of the candidates will not not find it preferable to manipulate. Thus, for any θ that is extreme enough, relative to the threshold $\hat{x}$, there is a unique pure-strategy equilibrium, where no candidate engages in population.

However, for θ close to $\hat{x}$, more precisely $\theta \in (\hat{x} - \frac{2}{\sqrt{k_I}}, \hat{x} + \frac{2}{\sqrt{k_C}})$, "no manipulation" ceases to be equilibrium strategy as at least one candidate prefer deviating to manipulation. We will show below that there is no pure-strategy simultaneous response by candidates for θ close to $\hat{x}$.

Firstly note that there is no equilibrium that both candidates engage in manipulation, since the vote shares cannot increase incrementally. Then, there is no equilibrium, where C engages in manipulation, since I can simply revert the manipulation of C with less cost.

Now, if I engages in manipulation so that there is a bijection between θ and $\theta + a_I(\theta)$, then voters can figure out θ from the public signal they receive. Thus manipulation would be obsolete in that case. This implies that if a pure-strategy equilibrium exists, then it has to include some pooling; that is, the candidate I maps, via manipulation, a non-singleton set of relative positions to single public signal. Moreover, for each such public signal, the pooled set of relative positions should include positions where both C is the better candidate and I is the better candidate, since otherwise voters would figure out, from the public signal, which candidate is better.

Now suppose that $\hat{\theta}$ is the minimal public signal that for any other public signal $\hat{\theta}'$

$< \hat{\theta}$ voters supoort C. Let $\underline{\Theta}$ and $\overline{\Theta}$ be the regions of relative positions that for any $\theta \in \underline{\Theta}$, $\overline{\Theta}$ $a_I(\theta) = \hat{\theta} - \theta$. Moreover, for any $\theta \in \underline{\Theta}$ $\theta \leq \hat{x}$ and for any $\theta \in \overline{\Theta}$ $\theta \geq \hat{x}$.

Firstly, for any $\theta \in \underline{\Theta}$, $\overline{\Theta}$, and any manipulative action $a \in [0,\ a_I(\theta))$, whenever voters observe $\theta + a$, they should support the candidate C, since otherwise I would prefer to reduce the manipulation level from $a_I(\theta))$ to a. Moreover since there cannot be negative manipulation, $\hat{\theta} \geq sup\overline{\Theta}$.

Suppose that $\hat{\theta}$ is strictly greater than $sup\overline{\Theta}$. Then if voters observe $\hat{\theta}' \in (sup\overline{\Theta},\ \hat{\theta})$ they vote for C, but then I would have the incentive to manipulate into $\hat{\theta}$. This would contradict the definition of $\overline{\Theta}$. Thus $\hat{\theta} = sup\overline{\Theta}$.

Now, we have a strategic circularity. Since for any public signal $\hat{\theta}' < \hat{\theta}$ voters support candidate C, then for any relative position $\theta < \hat{\theta}$ such that $|\theta - \hat{\theta}| < \frac{2}{\sqrt{k_C}}$, the candidate C could profitably manipulate. That is not an equilibrium either because I would prefer to revert the manipulation of C, which would force C to stop manipulation, then I would choose the manipulation level as above.

Thus, it is not possible to support a pure-strategy pooling equilibrium, where voters adopt monotonic strategies.

Mixed Strategy Equilibria

$\forall\ j \in P$, let γ_j be the mixed strategy that j plays, where $\gamma_j \in \Delta(\mathbb{R}_+)$ given the aggregate voter response function $\beta(\cdot)$. Now, given θ and $\gamma_C(\cdot)$, the expected utility of playing γ_I to I is

$$\int_0^\infty \int_0^\infty (1 - 2\beta(\theta + a_I - a_C) - H_I(a_I))\gamma_I(a_I)\gamma_C(a_C)da_I da_C,$$

and the expected utility of playing a degenerate strategy a_I is

$$\int_0^\infty (1 - 2\beta(\theta + a_I - a_C) - H_I(a_I))\gamma_C(a_C)da_C,$$

which is equal to

$$1 - H_I(a_I) - 2\int_{A_C} \beta(\theta + a_I - a_C)\gamma_C(a_C)da_C,$$

where A_C is the support of the mixed strategy of candidate C, $\gamma_C(\cdot)$

Given the beliefs of candidates in monotonic strategies of voters, the equation above amounts to

$$1 - H_I(a_I) - 2\int_{inf(A_C)}^{\theta+a_I-\hat{x}} \gamma_C(a_C)da_C.$$

Note that for any two pure strategies a, a' in the support of γ_I,

$$\begin{aligned} &1 - H_I(a) - 2\int_{inf(A_C)}^{\theta+a-\hat{x}} \gamma_C(a_C)da_C \\ &= 1 - H_I(a') - 2\int_{inf(A_C)}^{\theta+a'-\hat{x}} \gamma_C(a_C)da_C. \end{aligned}$$

However this equality implies by Assumption 1 that

$$\frac{1}{2}k_I(a'^2 - a^2) = 2\left(\int_{inf(A_C)}^{\theta+a-\hat{x}} \gamma_C(a_C)da_C - \int_{inf(A_C)}^{\theta+a'-\hat{x}} \gamma_C(a_C)da_C\right),$$

which holds only if $a = a'$. This shows that a mixed strategy is not a best-response for I. A similar argument shows that C does not play mixed strategies as well. Thus given that voters play monotonically, there are no mixed strategy equilibria either.

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